Cybersecurity Alert Summarizer

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Introduction

- **Problem:** SIEM and IDS tools raise thousands of near-identical alerts for a single attack, so analysts suffer alert fatigue and the real incident is easy to miss.
- **Data:** AIT Alert Data Set (Landauer et al.): real Wazuh + Suricata alerts with labelled attack phases; scenarios russellmitchell (41,488 alerts) and shaw (68,539 alerts).
- **Objectives:** group alerts into incidents; summarise each one (what, who, target, when, severity, next step); answer questions; run offline on a CPU.
- **Scope:** all five NLP units in a React + FastAPI + SQLite web app, with an NLP Lab that shows every step.

Methods

- **Alert → sentence:** each JSON alert becomes English, e.g. “At 09:40:06, host mail reported 'sshd: authentication success' for user jhall from 172.19.131.174 (level 3).”
- **Incidents:** TF-IDF + k-means (k chosen by silhouette), split at 30-minute gaps.
- **Summary:** template filling, with extractive and DistilBART baselines; a QA chatbot over the facts.
- **Own code:** edit distance, bigram LM, Viterbi, CKY, Chu-Liu/Edmonds, PMI.



Results

Key findings

- **Alerts → incidents:** russellmitchell 41,488 raw → 21,354 unique alerts → 214 incidents; shaw 68,539 → 30,732 → 444; bundled sample 1,515 → 1,467 → 18.
- **Purity:** incidents match the attack phases with purity 0.835 and 0.974, above the one-group baseline of 0.624 and 0.822 (Fig. 4).
- **Language model:** held-out perplexity is ∞ without smoothing and 8.31 / 11.30 with Laplace smoothing.
- **Faithful summaries:** template summaries keep IPs, times and counts exact; DistilBART once shortened an IP address.
- **Speed:** a full scenario runs in 37.3 s and 40.4 s on a laptop CPU.
- **Ask tab + PDF:** questions are answered with supporting alerts; each incident exports as a PDF report.

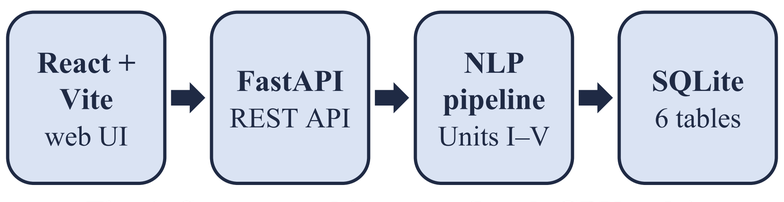


Fig. 1 System architecture (local, CPU only)

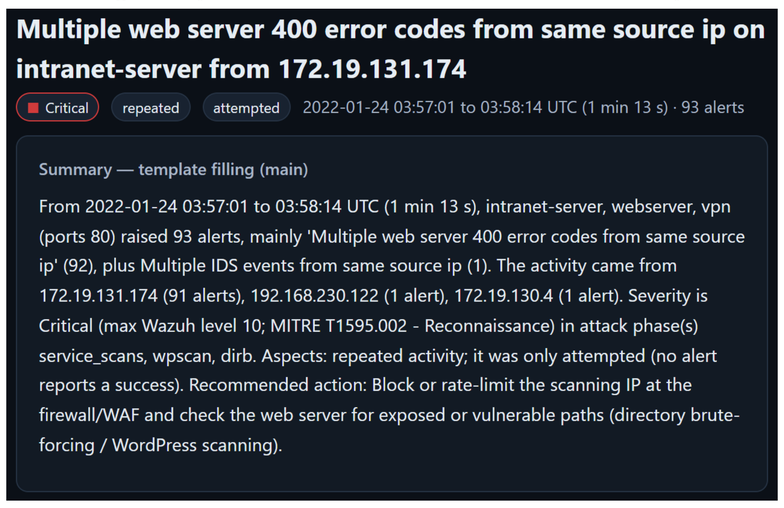


Fig. 2 Incident page with its template summary



Fig. 3 NLP Lab, Unit II: our CKY parse table

Summary quality (ROUGE F1)

Summary	R-1	R-2	R-L
Template	**0.4630**	**0.1804**	**0.3095**
Extractive	0.3293	0.1621	0.2456
DistilBART	0.3489	0.1661	0.2457

Mean F1 over 25 hand-written reference summaries; template row highlighted.

- **NER:** micro F1 0.994 on 50 sentences.
- **Triples:** precision = recall = 0.65 on 20 gold sentences.
- **HMM vs spaCy:** POS agreement 0.678 (our Viterbi).
- **Tests:** 19 pytest tests pass.

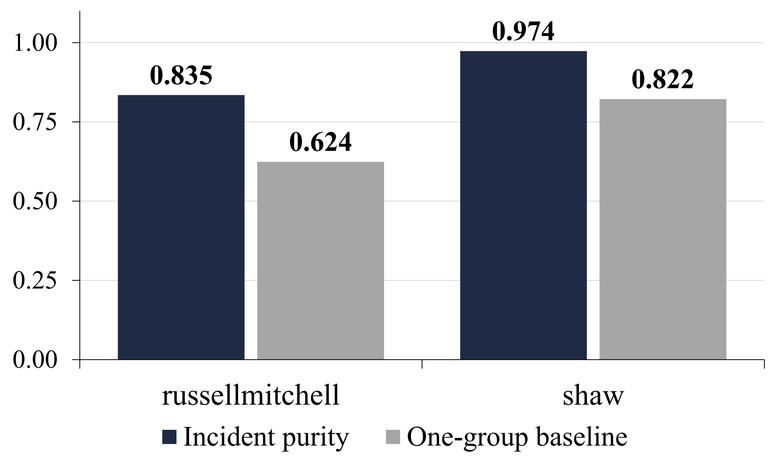


Fig. 4 Incident purity vs one-group baseline

Discussions

Conclusion

- A local, explainable pipeline turns tens of thousands of raw alerts into a few hundred incidents with summaries that keep the facts exact.
- Template filling beat the extractive and DistilBART summaries on every ROUGE score.

Limitations

- Gold data was labelled by one annotator.
- The NER score is optimistic: most test sentences follow our own template.
- Two attackers doing the same thing at the same time merge into one incident.

Future work

- Live alert streams instead of uploaded files.
- Attacker-aware clustering.
- Domain-trained models, plus translation and speech for Unit V.

Scan for the code and results

Dataset: Landauer et al., AIT Alert Data Set (AIT-ADS), Zenodo 8263181, CC BY 4.0. Code: github.com/Abin-Shaji-Thomas/Cybersecurity-Alert-Summarizer